

Integrated Geneformer Analysis Report

Combined T-cell and Epithelial Cell Analyses with Notebook Documentation

Integrated Geneformer Report: T-cell and Epithelial Analyses

Prepared for: medical collaborators, pathology collaborators, and translational research team
Datasets: lung single-cell RNA-seq dataset containing normal lung, LUAD, and SCLC samples
Analyses integrated: T-cell disease-state analysis and epithelial/tumor-cell disease-state analysis

1. Executive Summary

We performed two complementary Geneformer analyses on lung single-cell RNA-seq data:

1. T-cell analysis:

- Focused on disease-associated immune states across normal lung, LUAD, and SCLC.
- Geneformer detected disease-associated T-cell signal, but classification was only moderate.
- The model tended to classify many normal and SCLC T cells as LUAD, suggesting substantial overlap among tumor-associated immune programs.

2. Epithelial analysis:

- Focused on tumor epithelial cells from LUAD and SCLC.
- Geneformer strongly distinguished LUAD from SCLC epithelial cells.
- The epithelial classifier showed much stronger disease separation than the T-cell classifier.
- In silico perturbation identified candidate genes whose simulated overexpression shifted LUAD epithelial cells toward an SCLC-like state.

Overall conclusion:

The strongest disease-specific transcriptional signal is in epithelial tumor cells. T cells show disease-associated immune remodeling, but the T-cell states overlap substantially across LUAD and SCLC. This supports a biological model in which tumor epithelial identity drives lineage-specific disease separation, while the immune compartment reflects a more shared cancer-associated microenvironment.

2. Starting Dataset

The full lung cancer single-cell dataset contained approximately:

- 147,137 cells
- 22,021 genes

Major disease groups included:

- small cell lung carcinoma (SCLC)
- lung adenocarcinoma (LUAD)
- normal lung

The project analyzed two major compartments:

- T cells, representing the immune microenvironment
- epithelial cells, representing the malignant/tumor compartment

3. T-cell Geneformer Analysis

3.1 Biological question

Can T-cell transcriptomes distinguish normal lung, LUAD, and SCLC disease states?

This question is clinically relevant because T cells reflect tumor immune microenvironment remodeling and may relate to immune checkpoint inhibitor response.

3.2 Model setup

The T-cell classifier was trained to predict:

- normal
- lung adenocarcinoma
- small cell lung carcinoma

from T-cell transcriptomes.

Patient-level splitting was used to reduce donor leakage.

3.3 Key performance

The T-cell classifier achieved:

- Accuracy: approximately 0.661
- Macro-F1: approximately 0.499

The confusion matrix showed:

- LUAD T cells were most often classified correctly.
- Many SCLC T cells were misclassified as LUAD.
- Many normal T cells were also misclassified as LUAD.

3.4 Interpretation

The T-cell model detected disease-related immune signal, but disease separation was incomplete.

The dominant LUAD prediction pattern suggests either:

- class imbalance,
- shared tumor-associated immune programs across lung cancers,
- or insufficiently distinct T-cell states between LUAD and SCLC.

Clinically, this implies that T cells alone may not robustly identify tumor lineage, but they likely capture broader immune microenvironment remodeling.

4. Epithelial Geneformer Analysis

4.1 Biological question

Can epithelial tumor-cell transcriptomes distinguish LUAD from SCLC, and can perturbation analysis identify genes associated with LUAD-to-SCLC-like transition?

4.2 Data preparation

The epithelial analysis filtered the full dataset to lung-derived epithelial cells.

The main two-class epithelial subset included:

- LUAD epithelial cells

- SCLC epithelial cells

The reported epithelial subset contained:

- 31,258 epithelial cells
- 22,021 genes

Disease composition:

- LUAD epithelial cells: 5,270 cells from 16 donors
- SCLC epithelial cells: 25,988 cells from 9 donors

4.3 Geneformer tokenization

The epithelial subset was tokenized using Geneformer V1. Metadata retained for downstream analysis included:

- cell type
- disease
- donor/individual identifier
- sex
- age
- tissue

4.4 LUAD vs SCLC epithelial classifier

The main epithelial classifier predicted:

- lung adenocarcinoma
- small cell lung carcinoma

The notebook reported strong validation performance:

- Validation accuracy: 0.951
- Validation macro-F1: 0.941
- Validation loss: 0.221
- Training loss: 0.0395

4.5 Interpretation

This strong performance indicates that Geneformer learned robust disease-specific epithelial transcriptional programs. This is biologically plausible because LUAD and SCLC differ strongly in tumor lineage, differentiation state, and neuroendocrine identity.

Compared with the T-cell classifier, epithelial cells contained much stronger disease-lineage information.

5. Exploratory Three-Class Epithelial Classifier

The notebook also trained a classifier including:

- normal epithelial cells
- LUAD epithelial cells
- SCLC epithelial cells

Reported validation performance:

- Validation accuracy: 0.850
- Validation macro-F1: 0.747

This was lower than the LUAD/SCLC two-class classifier, which is expected because the three-class problem is more complex and normal epithelial samples were less abundant.

Important caveat:

The three-class epithelial split appeared to have possible donor overlap/leakage in the notebook output. Therefore, the three-class epithelial result should be considered exploratory until rerun with strict donor-exclusive splitting.

6. Epithelial Embedding Analysis

Fine-tuned Geneformer embeddings were extracted from the LUAD/SCLC epithelial classifier.

Embedding output:

- 31,258 epithelial cells
- 258 embedding columns

These embeddings represent each epithelial cell in a learned disease-aware latent space. They provide the basis for visualization and perturbation experiments.

7. Epithelial In Silico Perturbation

7.1 Biological question

Which gene overexpression perturbations shift LUAD epithelial cells toward an SCLC-like transcriptional state?

This is relevant for studying:

- LUAD-to-SCLC transformation,
- tumor lineage plasticity,
- neuroendocrine transition,
- therapy resistance mechanisms.

7.2 Perturbation setup

Transition modeled:

- Start state: lung adenocarcinoma
- Goal state: small cell lung carcinoma
- Perturbation type: overexpression
- Genes perturbed: all detected genes
- Model: fine-tuned Geneformer epithelial classifier
- Max cells: 2,500

7.3 Perturbation output

The perturbation result table included:

- 15,531 genes
- shift toward the SCLC goal state
- FDR
- number of detections
- significance flag

After filtering:

- Goal_end_FDR < 0.05
- N_Detections > 100

the notebook reported 1,059 significant, sufficiently detected genes.

7.4 Example positive-shift genes

Genes whose overexpression shifted LUAD cells toward the SCLC goal state included:

- SFTPC
- TPT1
- SLPI
- PTMA
- ACTB

These should be interpreted cautiously. Some are epithelial-lineage or secretory markers, while others are broadly expressed housekeeping/proliferation-associated genes.

7.5 Example negative-shift genes

Genes with negative shifts included:

- B2M
- S100A9
- FTH1
- RALGAPA2
- ITGB8

Negative shift means simulated overexpression moved LUAD epithelial cells away from the SCLC goal state in the Geneformer latent space.

8. Integrated Interpretation

8.1 Cross-compartment comparison

T-cell compartment:

- Moderate disease classification
- Strong overlap between tumor-associated immune states
- Useful for studying immune remodeling and ICI biology

Epithelial compartment:

- Strong LUAD vs SCLC separation
- Strong tumor-lineage signal
- More suitable for studying lineage plasticity and LUAD-to-SCLC transition

8.2 Biological model

The analyses support the following model:

Tumor epithelial cells carry the primary disease-lineage signal.

T cells reflect the downstream immune response to cancer and show shared tumor-associated programs across LUAD and SCLC.

This means:

- epithelial Geneformer analysis is better for tumor identity and lineage transition,
- T-cell Geneformer analysis is better for immune microenvironment and immunotherapy-related hypotheses.

9. Clinical and Translational Relevance

For medical collaborators:

- Geneformer can separate LUAD and SCLC very well using epithelial tumor-cell transcriptomes.

- T-cell states carry disease information but are less specific to tumor lineage.
- Perturbation analysis can nominate candidate genes linked to tumor-state transitions, but these are computational predictions.

For pathology collaborators:

- The epithelial classifier supports strong transcriptional separation between LUAD and SCLC tumor cells.
- This aligns with the known biological distinction between glandular epithelial differentiation and neuroendocrine SCLC lineage.
- Perturbation results may help prioritize markers for further immunohistochemistry, spatial transcriptomics, or validation in independent datasets.

10. Limitations

Major limitations:

1. Geneformer V1 was used in the epithelial notebook; future work should compare V2 104M.
2. The epithelial LUAD/SCLC subset was imbalanced by both cell count and donor count.
3. Test metrics were not consistently saved in the notebook outputs.
4. The exploratory three-class epithelial classifier may have donor leakage.
5. Perturbation results are computational predictions and not direct experimental evidence.
6. Broadly expressed genes may appear as significant perturbation hits because of high detection frequency.
7. External validation datasets were not yet included.

11. Recommended Next Steps

Highest-priority technical steps:

1. Re-run both T-cell and epithelial analyses using a standardized run folder structure.
2. Apply strict donor-exclusive train/eval/test splits.
3. Save all metrics, confusion matrices, and prediction plots as CSV/JSON/PDF.
4. Repeat training across multiple random seeds.
5. Compare Geneformer V1 and V2.

Highest-priority biological steps:

1. For epithelial analysis, focus on known SCLC lineage factors:

- ASCL1
- NEUROD1
- POU2F3
- YAP1
- MYC family genes
- NOTCH pathway genes
- RB1 and TP53 pathway-associated programs

2. For T-cell analysis, focus on immune checkpoint and exhaustion programs:

- PDCD1
- CTLA4
- TIGIT
- LAG3
- IFNG
- TNF
- CXCL13
- TOX

3. Perform pathway enrichment on positive- and negative-shift perturbation genes separately.

4. Validate top candidates using:
- independent single-cell datasets,
 - spatial transcriptomics,
 - immunohistochemistry,
 - perturbation experiments,
 - known LUAD-to-SCLC transformation cohorts.

12. Suggested Collaborator Summary

We used Geneformer to analyze two compartments from lung cancer single-cell RNA-seq data: T cells and epithelial tumor cells. T-cell classification across normal lung, LUAD, and SCLC showed moderate performance, suggesting that immune cells carry disease-associated signals but also share common cancer-associated programs across tumor types. In contrast, epithelial tumor cells showed strong LUAD-vs-SCLC separation, with validation accuracy around 95% and macro-F1 around 94%. This indicates that tumor epithelial cells encode the strongest disease-lineage signal. In silico perturbation of epithelial cells identified candidate genes whose simulated overexpression shifts LUAD cells toward an SCLC-like state, nominating possible regulators of lineage plasticity and transformation. These results are preliminary and hypothesis-generating, but they support Geneformer as a useful framework for integrating tumor-cell identity, immune microenvironment remodeling, and computational perturbation in lung cancer.

Included Resources

1. Commented epithelial notebook generated from the original analysis notebook.
2. Integrated interpretation of T-cell and epithelial Geneformer findings.
3. Recommendations for validation, perturbation follow-up, and future analyses.